



University of Vavuniya

Second Examination in Information Technology - 2023

Second Semester - July 2025

IT2244 Operating System (Practical)

Answer all the questions

Time Allowed : Three hours

Instructions:

- Save all your answers corresponding to the question number in a folder named with your **index number**.

1. Write a bash script to organize a folder of image files. These image files are named using dates and times (e.g., 2025-07-16_14-30.jpg). The script should:

- (a) List all image files (with .jpg or .png extensions).
- (b) Create folders based on **year and month** (e.g., 2025-07).
- (c) Move images into the correct folders using the filename or metadata.
- (d) List how many images are in each folder.
- (e) Detect and delete duplicate images:

- *Hint: Use file size or checksum (e.g., md5sum) to detect duplicates.*

- (f) Generate a summary report showing how many files were organised.
- (g) Exit at the end.

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2. Write a C program to simulate a multi-level task manager. Use only `fork()`, `wait()`, and `sleep()` system calls.

(a) Level 0 – Main Task Manager

- This is the main parent process.
- Print: Task Name = "Main Task Manager", PID, PPID, and Level.

(b) Level 1 – Create 3 child processes to complete the tasks

- File Backup Task
- Network Monitoring Task
- Log Cleanup Task
- Each prints its task name, PID, PPID, and level.

(c) Level 2 – Subtasks

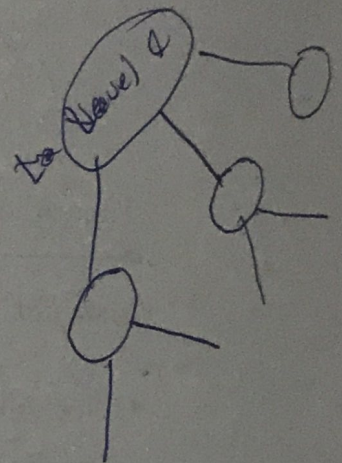
- File Backup Task creates:
 - Read Files
 - Write Backup
- Network Monitoring Task creates:
 - Capture Packets
 - Analyse Traffic

(d) Log Cleanup Task has no subtasks.

(e) Use `sleep(1)` or `sleep(2)` before or after printing each task to simulate that tasks are running.

(f) Use `wait()` to synchronize:

- Main Task Manager waits for all Level 1 tasks.
- Level 1 tasks must wait for their Level 2 subtasks.



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3. Write a C program to simulate patient vital (heart rate, oxygen level) monitoring using pipe() system call.

(a) Collect patient data: name, heart rate, oxygen level.

(b) Use pipe() to simulate internal communication between:

- Monitoring Unit (writes data)
- Decision Unit (reads and analyses data)

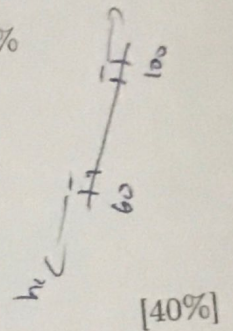
(c) Check vital signs and classify the status as:

- **CRITICAL:** (heart rate < 60 or heart rate > 100) OR oxygen < 90%
- **ALERT:** heart rate 60-65 OR 95-100 AND oxygen 90-92%
- **STABLE:** heart rate 60-100 AND oxygen > 92%

(d) Print patient report with status.

(e) Count total number of critical cases.

heart-rate
oxygen

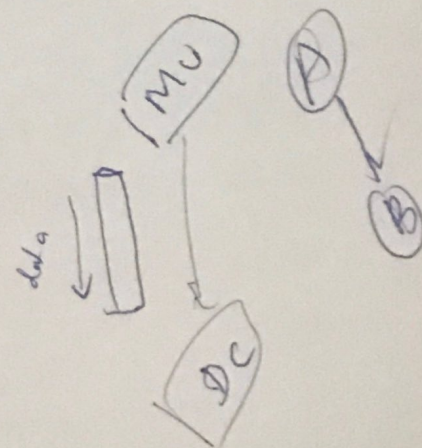


Sample Output:

===== Hospital Patient Monitoring System =====

Total Patients Monitored: 5

Patient Name	Heart Rate	Oxygen Level	Status
John Doe	85	98%	STABLE
Jane Smith	45	88%	CRITICAL
Alex Roy	97	90%	ALERT
Emily Davis	76	85%	CRITICAL
Chris Green	65	91%	ALERT



Patients in Critical Condition: 2